- amount of discount = discount rate · original price
- sale price = original price − discount

- **Mark-up** The mark-up is the amount added to the wholesale price, determined by the mark-up rate.
  - amount of mark-up = mark-up rate wholesale price
  - list price = wholesale price + mark up

## 6.4 Solve Simple Interest Applications

- **Simple interest**
  - If an amount of money, $P$, the principal, is invested for a period of $t$ years at an annual interest rate $r$, the amount of interest, $I$, earned is $I = Prt$
  - Interest earned according to this formula is called **simple interest**.

## 6.5 Solve Proportions and their Applications

- **Proportion**
  - A proportion is an equation of the form $\frac{a}{b} = \frac{c}{d}$, where $b \neq 0$, $d \neq 0$.The proportion states two ratios or rates are equal. The proportion is read "$a$ is to $b$, as $c$ is to $d$".

- **Cross Products of a Proportion**
  - For any proportion of the form $\frac{a}{b} = \frac{c}{d}$, where $b \neq 0$, its cross products are equal: $a \cdot d = b \cdot c$.

- **Percent Proportion**
  - The amount is to the base as the percent is to 100. $\frac{\text{amount}}{\text{base}} = \frac{\text{percent}}{100}$

# Exercises

## Review Exercises

### Understand Percent

*In the following exercises, write each percent as a ratio.*

**331**. 32% admission rate for the university

**332**. 53.3% rate of college students with student loans

*In the following exercises, write as a ratio and then as a percent.*

**333**. 13 out of 100 architects are women.

**334**. 9 out of every 100 nurses are men.

*In the following exercises, convert each percent to a fraction.*

**335**. 48%

**336**. 175%

**337**. 64.1%

**338**. $8\frac{1}{4}\%$

*In the following exercises, convert each percent to a decimal.*

**339**. 6%

**340**. 23%

**341**. 128%

**342**. 4.9%

*In the following exercises, convert each percent to ⓐ a simplified fraction and ⓑ a decimal.*

**343**. In 2012, 13.5% of the United States population was age 65 or over. (Source: www.census.gov)

**344**. In 2012, 6.5% of the United States population was under 5 years old. (Source: www.census.gov)

**345**. When a die is tossed, the probability it will land with an even number of dots on the top side is 50%.

**346**. A couple plans to have three children. The probability they will all be girls is 12.5%.

*In the following exercises, convert each decimal to a percent.*

**347**. $0.04$

**348**. $0.15$

**349**. $2.82$

**350**. $3$

**351**. $0.003$

**352**. $1.395$

*In the following exercises, convert each fraction to a percent.*

**353**. $\frac{3}{4}$

**354**. $\frac{11}{5}$

**355**. $3\frac{5}{8}$

**356**. $\frac{2}{9}$

**357**. According to the Centers for Disease Control, $\frac{2}{5}$ of adults do not take a vitamin or supplement.

**358**. According to the Centers for Disease Control, among adults who do take a vitamin or supplement, $\frac{3}{4}$ take a multivitamin.

*In the following exercises, translate and solve.*

**359**. What number is $46\%$ of $350$?

**360**. $120\%$ of $55$ is what number?

**361**. $84$ is $35\%$ of what number?

**362**. $15$ is $8\%$ of what number?

**363**. $200\%$ of what number is $50$?

**364**. $7.9\%$ of what number is $\$4.74$?

**365**. What percent of $120$ is $81.6$?

**366**. What percent of $340$ is $595$?

## Solve General Applications of Percents

*In the following exercises, solve.*

**367**. When Aurelio and his family ate dinner at a restaurant, the bill was $\$83.50$. Aurelio wants to leave $20\%$ of the total bill as a tip. How much should the tip be?

**368**. One granola bar has $2$ grams of fiber, which is $8\%$ of the recommended daily amount. What is the total recommended daily amount of fiber?

**369**. The nutrition label on a package of granola bars says that each granola bar has $190$ calories, and $54$ calories are from fat. What percent of the total calories is from fat?

**370**. Elsa gets paid $\$4{,}600$ per month. Her car payment is $\$253$. What percent of her monthly pay goes to her car payment?

*In the following exercises, solve.*

**371**. Jorge got a raise in his hourly pay, from $\$19.00$ to $\$19.76$. Find the percent increase.

**372**. Last year Bernard bought a new car for $\$30{,}000$. This year the car is worth $\$24{,}000$. Find the percent decrease.

## Solve Sales Tax, Commission, and Discount Applications

*In the following exercises, find ⓐ the sales tax ⓑ the total cost.*

**373**. The cost of a lawn mower was $\$750$. The sales tax rate is $6\%$ of the purchase price.

**374**. The cost of a water heater is $\$577$. The sales tax rate is $8.75\%$ of the purchase price.

*In the following exercises, find the sales tax rate.*

**375**. Andy bought a piano for $4,600. The sales tax on the purchase was $333.50.

**376**. Nahomi bought a purse for $200. The sales tax on the purchase was $16.75.

*In the following exercises, find the commission.*

**377**. Ginny is a realtor. She receives 3% commission when she sells a house. How much commission will she receive for selling a house for $380,000?

**378**. Jackson receives 16.5% commission when he sells a dinette set. How much commission will he receive for selling a dinette set for $895?

*In the following exercises, find the rate of commission.*

**379**. Ruben received $675 commission when he sold a $4,500 painting at the art gallery where he works. What was the rate of commission?

**380**. Tori received $80.75 for selling a $950 membership at her gym. What was her rate of commission?

*In the following exercises, find the sale price.*

**381**. Aya bought a pair of shoes that was on sale for $30 off. The original price of the shoes was $75.

**382**. Takwanna saw a cookware set she liked on sale for $145 off. The original price of the cookware was $312.

*In the following exercises, find ⓐ the amount of discount and ⓑ the sale price.*

**383**. Nga bought a microwave for her office. The microwave was discounted 30% from an original price of $84.90.

**384**. Jarrett bought a tie that was discounted 65% from an original price of $45.

*In the following exercises, find ⓐ the amount of discount ⓑ the discount rate. (Round to the nearest tenth of a percent if needed.)*

**385**. Hilda bought a bedspread on sale for $37. The original price of the bedspread was $50.

**386**. Tyler bought a phone on sale for $49.99. The original price of the phone was $79.99.

*In the following exercises, find*

ⓐ the amount of the mark-up ⓑ the list price

**387**. Manny paid $0.80 a pound for apples. He added 60% mark-up before selling them at his produce stand. What price did he charge for the apples?

**388**. It cost Noelle $17.40 for the materials she used to make a purse. She added a 325% mark-up before selling it at her friend's store. What price did she ask for the purse?

### Solve Simple Interest Applications

*In the following exercises, solve the simple interest problem.*

**389**. Find the simple interest earned after 4 years on $2,250 invested at an interest rate of 5%.

**390**. Find the simple interest earned after 7 years on $12,000 invested at an interest rate of 8.5%.

**391**. Find the principal invested if $660 interest was earned in 5 years at an interest rate of 3%.

**392**. Find the interest rate if $2,898 interest was earned from a principal of $23,000 invested for 3 years.

**393**. Kazuo deposited $10,000 in a bank account with interest rate 4.5%. How much interest was earned in 2 years?

**394**. Brent invested $23,000 in a friend's business. In 5 years the friend paid him the $23,000 plus $9,200 interest. What was the rate of interest?

**395**. Fresia lent her son $5,000 for college expenses. Three years later he repaid her the $5,000 plus $375 interest. What was the rate of interest?

**396**. In 6 years, a bond that paid 5.5% earned $594 interest. What was the principal of the bond?

### Solve Proportions and their Applications

*In the following exercises, write each sentence as a proportion.*

**397**. 3 is to 8 as 12 is to 32.

**398**. 95 miles to 3 gallons is the same as 475 miles to 15 gallons.

**399**. 1 teacher to 18 students is the same as 23 teachers to 414 students.

**400**. $7.35 for 15 ounces is the same as $2.94 for 6 ounces.

*In the following exercises, determine whether each equation is a proportion.*

**401**. $\frac{5}{13} = \frac{30}{78}$

**402**. $\frac{16}{7} = \frac{48}{23}$

**403**. $\frac{12}{18} = \frac{6.99}{10.99}$

**404**. $\frac{11.6}{9.2} = \frac{37.12}{29.44}$

*In the following exercises, solve each proportion.*

**405**. $\frac{x}{36} = \frac{5}{9}$

**406**. $\frac{7}{a} = \frac{-6}{84}$

**407**. $\frac{1.2}{1.8} = \frac{d}{6}$

**408**. $\frac{\frac{1}{2}}{2} = \frac{m}{20}$

*In the following exercises, solve the proportion problem.*

**409**. The children's dosage of acetaminophen is $5$ milliliters (ml) for every $25$ pounds of a child's weight. How many milliliters of acetaminophen will be prescribed for a $60$ pound child?

**410**. After a workout, Dennis takes his pulse for $10$ sec and counts $21$ beats. How many beats per minute is this?

**411**. An $8$ ounce serving of ice cream has $272$ calories. If Lavonne eats $10$ ounces of ice cream, how many calories does she get?

**412**. Alma is going to Europe and wants to exchange $\$1{,}200$ into Euros. If each dollar is $0.75$ Euros, how many Euros will Alma get?

**413**. Zack wants to drive from Omaha to Denver, a distance of $494$ miles. If his car gets $38$ miles to the gallon, how many gallons of gas will Zack need to get to Denver?

**414**. Teresa is planning a party for $100$ people. Each gallon of punch will serve $18$ people. How many gallons of punch will she need?

*In the following exercises, translate to a proportion.*

**415**. What number is $62\%$ of $395$?

**416**. $42$ is $70\%$ of what number?

**417**. What percent of $1{,}000$ is $15$?

**418**. What percent of $140$ is $210$?

*In the following exercises, translate and solve using proportions.*

**419**. What number is $85\%$ of $900$?

**420**. $6\%$ of what number is $\$24$?

**421**. $\$3.51$ is $4.5\%$ of what number?

**422**. What percent of $3{,}100$ is $930$?

## Practice Test

*In the following exercises, convert each percent to ⓐ a decimal ⓑ a simplified fraction.*

**423**. $24\%$

**424**. $5\%$

**425**. $350\%$

*In the following exercises, convert each fraction to a percent. (Round to $3$ decimal places if needed.)*

**426**. $\frac{7}{8}$

**427**. $\frac{1}{3}$

**428**. $\frac{11}{12}$

*In the following exercises, solve the percent problem.*

**429**. $65$ is what percent of $260$?

**430**. What number is $27\%$ of $3{,}000$?

**431**. $150\%$ of what number is $60$?

**432**. Yuki's monthly paycheck is $\$3{,}825$. She pays $\$918$ for rent. What percent of her paycheck goes to rent?

**433**. The total number of vehicles on one freeway dropped from $84{,}000$ to $74{,}000$. Find the percent decrease (round to the nearest tenth of a percent).

**434**. Kyle bought a bicycle in Denver where the sales tax was $7.72\%$ of the purchase price. The purchase price of the bicycle was $\$600$. What was the total cost?

**435**. Mara received $\$31.80$ commission when she sold a $\$795$ suit. What was her rate of commission?

**436**. Kiyoshi bought a television set on sale for $\$899$. The original price was $\$1{,}200$. Find:
ⓐ the amount of discount
ⓑ the discount rate (round to the nearest tenth of a percent)

**437**. Oxana bought a dresser at a garage sale for $\$20$. She refinished it, then added a $250\%$ markup before advertising it for sale. What price did she ask for the dresser?

**438**. Find the simple interest earned after $5$ years on $\$3000$ invested at an interest rate of $4.2\%$.

**439**. Brenda borrowed $\$400$ from her brother. Two years later, she repaid the $\$400$ plus $\$50$ interest. What was the rate of interest?

**440**. Write as a proportion: $4$ gallons to $144$ miles is the same as $10$ gallons to $360$ miles.

**441**. Solve for a: $\frac{12}{a} = \frac{-15}{65}$

**442**. Vin read $10$ pages of a book in $12$ minutes. At that rate, how long will it take him to read $35$ pages?